Tan Kiet HONG

Civil & Geomechanics Engineer

Email: hogtankit2001@gmail.com — Mobile: (+33) 7 46 50 21 91
GitHub: tankiet1907 — LinkedIn: tankiethong — Address: Grenoble, France

EDUCATION

Université Grenoble Alpes **Sep 2025 - Jul 2026**

Master 2: Geomechanics, Civil Engineering & Risks *Grenoble, France*

- **M2 Final Grade:** 14.235/20 (Admis - Mention Bien) | **Semester 1:** 13.47/20 | **Semester 2:** 15.00/20.
- **Key Coursework:** Numerical Methods for Nonlinear Mechanics, Durability and Vulnerability of Structures (15.8/20), Quantitative Image Analysis for Mechanics (16/20).
- **Thesis:** Numerical study of the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a stochastic poro-mechanical approach (15/20).

Ho Chi Minh University of Technology **Sep 2019 - Jun 2024**

Engineering Degree in Civil Engineering *Ho Chi Minh City, Vietnam*

(Specialty: Building & Energy)

- **GPA:** 8.06/10 (3.5/4.0) | **Rank:** 3/12 (PFIEV Excellence Program). Graduation classification: Very Good.
- **Honors:** University Semester Scholarships for 4 consecutive years (2021-2024).
- **Thesis:** Circus And Multifunction Performance House Of HCM City. **Grade:** 8.10/10.

RESEARCH & PROFESSIONAL EXPERIENCE

M2 Research Intern **Feb 2026 - Jun 2026**

3SR Laboratory, Université Grenoble Alpes *Grenoble, France*

Supervisors: Prof. Julien Baroth (julien.baroth@univ-grenoble-alpes.fr) & Prof. Stefano Dal Pont (stefano.dalpont@univ-grenoble-alpes.fr)
Repository: github.com/tankiet1907/cast3m-m2internship

- Implemented and calibrated a non-linear Thermo-Hydro-Chemical (THC) model of concrete in Cast3M to analyze explosive spalling mechanisms.
- Developed an automated parameter sweep and data extraction tool in Python to interface with Cast3M for independent thermal and hydric calibrations.
- Conducted probabilistic sensitivity analysis (Sobol indices) and reliability assessments (probability of failure) using OpenTURNS with PCE metamodels.
- Modeled spatial material heterogeneity via 3D random fields using the Turning Bands Method (ALEA in Cast3M) to evaluate localized cracking risks.

Structure Engineer (Full Time) **Aug 2024 - Aug 2025**
GBC Engineers Vietnam *Ho Chi Minh City, Vietnam*

- Prepared Finite Element Models (FEM) and structural models for rigorous mechanical analysis.
- Delivered structural calculations, analysis reports, and correspondence ensuring compliance with international standards (Eurocode, TCVN).

Site Engineer (Internship) **Jun 2023 - Aug 2023**
Unicons Construction (Member of Coteccons Group) *Binh Duong, Vietnam*

- **Project:** LEGO Manufacturing Vietnam (Phase 1) – Moulding Building.
- Supervised structural fire protection implementations (Fire Protection Boards, Intumescent Paint, Concrete-encasement) to meet rigorous 120-minute Fire Protection Ratings.
- Optimized the execution schedule by transitioning the concrete encasement casting for columns from vertical to horizontal pouring.
- Conducted Quality Control per EC2: inspected steel frame erection via total station, verified tightening forces for Snug and Bearing bolts, and supervised Sika grouting and composite decking.

PROJECTS

Scientific Research: Fire-Resistant Steel Structures **Sep 2023 - Dec 2023**
Ho Chi Minh City University of Technology *Ho Chi Minh City, Vietnam*

- Evaluated thermo-mechanical properties of steel at high temperatures per EUROCODE specifications.
- Designed and verified fire-resistant structural elements (columns and beams) using advanced material models.

CERTIFICATIONS & AWARDS

- **Third Prize,** 36^{th} **Loa Thanh Award (2024):** National award for the most prestigious graduation projects in civil engineering.
- **Third Prize in Physics (2018):** City Level Excellent Student Competition (Ho Chi Minh City).
- **Certificate of Merit:** Recognized as a "Very Good Student" for the 2021-2022 academic year.

SKILLS & COMPETENCIES

Technical: Finite Element Analysis (FEA), THM/THC Multiphysics Coupling, Probabilistic Sensitivity and Reliability Analysis, Random Fields.

Software: Cast3M (Advanced), OpenTURNS, Python, LaTeX, VS Code, Linux, Revit Structure, AutoCAD, ETABS, SAP2000, MS Office.

Languages: English (Advanced - TOEIC: 935/990), French (Intermediate - TCF TP: 347/699, B1), Vietnamese (Native).